

Final Report

Andrew Nguyen

University of Michigan - School of Information

SI 305: Introduction to Information Analysis

Professor Ben Green, Kelvin Chang

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I. Summary

This project examines the effectiveness of Lead Safe Lansing's (LSL) in targeting ZIP codes with high percentages of children under 6 with elevated blood lead levels (EBLL), which are moderately associated with poverty. While this initiative identified and addressed some of the highest-risk areas, average blood lead levels remain higher than in neighborhoods less focused by the intervention. Also, the number of projects declined from 2010 to 2022, which might have reflected inflation. The paper will also examine the relationship between project count and inflation, as measured by the Producer Price Index (PPI), and find plausible evidence of an association after filtering out outliers.

II. Problem Statement

In 2024, 140,498 children under age 6 in Michigan were tested for elevated blood lead levels (EBLLs), of whom 5,846 (4.2%) had EBLLs of 3.5 µg/dL or higher.¹ This indicates that lead exposure continues to pose a significant public health concern across the state. EBLLs are linked to cognitive deficits, behavioral issues, and cardiovascular problems, such as increased blood pressure, highlighting the critical need for continued prevention and remediation efforts. This situation underscores the need to focus more on addressing these challenges.

Focusing on the city of Lansing, lead poisoning remains a significant concern. The Ingham County Health Department has reported that 70% of lead-poisoned children in the county reside in rental housing.² Despite LSL's ongoing efforts to address this issue, it remains a persistent problem that adversely affects young children. A major contributing factor is that children from low-income households are disproportionately exposed to lead risk, primarily because they are concentrated in older, pre-1978 rental properties located in historically underinvested neighborhoods, where deteriorating lead-based paint and contaminated dust present the primary pathways for exposure.³ Guest lectures by

¹ Lanphear, Bruce, Ana Navas-Acien, and David C. Bellinger. "Lead Poisoning." *New England Journal of Medicine* 391, no. 17 (2024): 1621–31. <https://doi.org/10.1056/NEJMr2402527>.

² Ingham County Health Department, "Lead Poisoning," accessed April 26, 2026, https://dei.ingham.org/environmental_health/lead_poisoning.php.

³ "Child Lead Exposure Elimination Commission," n.d., <https://www.michigan.gov/mdhhs/doing-business/commissions-boards/childhoodlead/child-lead-exposure-elimination-commission>.

LSL staff members Erin Buitendorp and Tyler Blakey highlight that these risks are particularly severe for children under 6 years old and pregnant women, emphasizing the urgent need to prioritize high-risk households in local abatement initiatives.⁴

LSL has been conducting abatement efforts since 2004 (the earliest completed year in their dataset), but they have not been able to effectively collect information, a glaring structural gap in the program's data-collection framework that prevents it from assessing whether resources are reaching the most vulnerable residents. LSL's administrative records are grouped by property and project, not by the residents who live within them. And the program doesn't reliably collect data on tenant income, length of tenancy, housing stability, or neighborhood-level socioeconomic indicators, such as poverty rates.⁵ Without this information at the resident level, city officials, program staff, and state agencies can't tell whether abatement resources are reaching low-income renters, those least able to self-finance remediation, and most at risk of displacement after improvements to their property. That complicates the ability to evaluate if wealthier property owners are reaping the largest share of program benefits, raising deep equity questions about how a publicly funded program spends its scarce resources in the name of economic progress.

III. Research Question

Research Question 1: Among Lansing, Michigan ZIP codes from 2010 to 2022, is Lead Safe Lansing's greater project concentration in higher-poverty, higher-EBLL areas associated with reductions in the percentage of children under age 6 with EBLLs?

Research Question 2: For Lead Safe Lansing residential abatement projects completed in Lansing, Michigan, between 2010 and 2022, how are rehabilitation costs related to changes in general price inflation over time?

⁴ Erin Buitendorp, "L05: Erin Buitendorp Visit" (Lansing, United States of America, January 26, 2025)

⁵ Lead Safe Lansing, "ProjectMasterExport_4.24.25 Raw Data_Single Family and Rental Abatement," Data set, April 25, 2025, https://docs.google.com/spreadsheets/d/1uv0w4QM1ayrjRXjiUeEEigzn_cmqnZd/edit?usp=drive_link&ouid=111817903337851474930&rtopof=true&sd=true.

IV. Analysis for RQ1:

A. Methodology:

To construct a longitudinal dataset, I merged three sources by ZIP code: Michigan's EBLI data for children younger than 6 years of age, socioeconomic data from the National Neighborhood Data Archive (NaNDA), and LSL's administrative project records, 2010–2022. NaNDA's poverty measure (PPOV) was constructed by combining 2010 and 2020 census geographies (ZCTA10, ZCTA20) for complete timespan coverage. I merged the dataset by ZIP code to compute mean EBLI rates and mean poverty proportions, and separately identified Lansing's ZIP codes and LSL's existence. LSL project counts were categorized into quintiles to capture variation in regional intensity. For the scatterplot analysis, I applied two robustness checks requiring at least 10 tested children per ZIP code to ensure statistical reliability and removed outliers using the interquartile range (IQR) method. I then ran Pearson's correlation coefficient for a high-level measure of association, and weighted least squares (WLS) with HC3 heteroscedasticity-robust standard errors as a conservative alternative that accounts for differences in ZIP code sample sizes. For the line plot, I ran a one-way ANOVA across the five quintile groups to test for structural differences in EBLI trends, followed by a Tukey HSD test to identify a quintile pair with statistically significant differences.

B. Finding & Interpretation

1. *Finding:*

The scatterplot indicates a positive correlation between poverty proportion and the proportions of children with EBLI. The Pearson's correlation coefficient and the WLS test show a 0.40 coefficient with a p-value below 0.000, indicating that we reject the null hypothesis of no relationship. We conclude that below-poverty income proportions relative to EBLI rates have a moderate and positive relationship affecting children under 6 years of age. For the line plot, ANOVA testing for structural differences across the five quintiles yields an F-statistic of 6.4 and a p-value of 0.0002, suggesting a statistically significant difference unlikely to be due to random error. A follow-up Tukey HSD test revealed three significant pairs ($p < 0.05$): Q1-Q3, Q1-Q4, and

Q4-Q5, and that Q4 is observed in two of the three pairs, indicating that the 4th quintile consistently demonstrates higher EBLL than the other groups.

2. Interpretation:

The moderate poverty-EBLL relationship indicates that LSL is typically reaching the right areas; the only ZIP code without project coverage has a relatively low poverty rate and EBLL percentage, suggesting that resources are not being wasted in low-risk areas. But the line plot gives a more nuanced picture. Although EBLL declines over time across all five quintile groups, Q4 remains persistently elevated throughout the period, whereas Q5, the highest project-concentration group, shows a steep decline. This is consistent with a dose-response relationship, in which significant EBLL reductions occur only at the highest levels of program intensity. This indicates moderate investment, such as in Q4, may not be enough to produce a change in health status, and that LSL's resources at present leave meaningful resource gaps in mid-concentration areas.

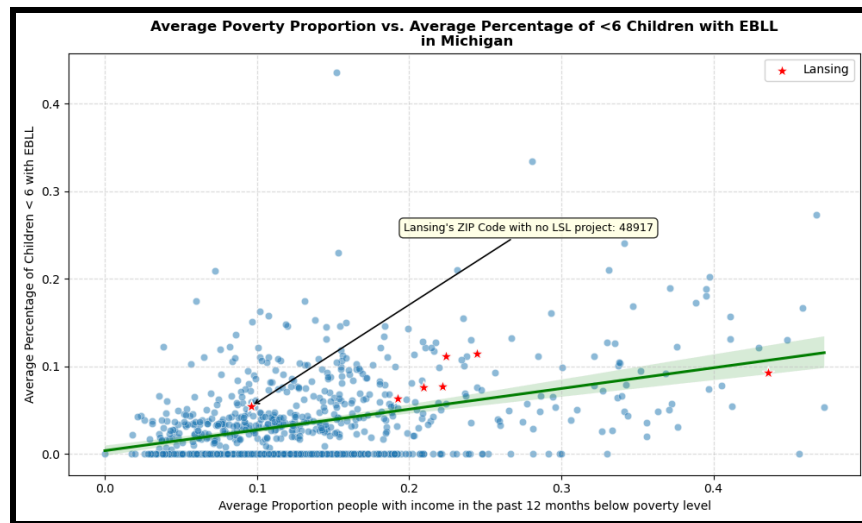


Figure 1. Average EBLL tends to increase when the Average Proportion of People with income below the poverty level increases, indicating worsened poisoning in more disadvantaged regions

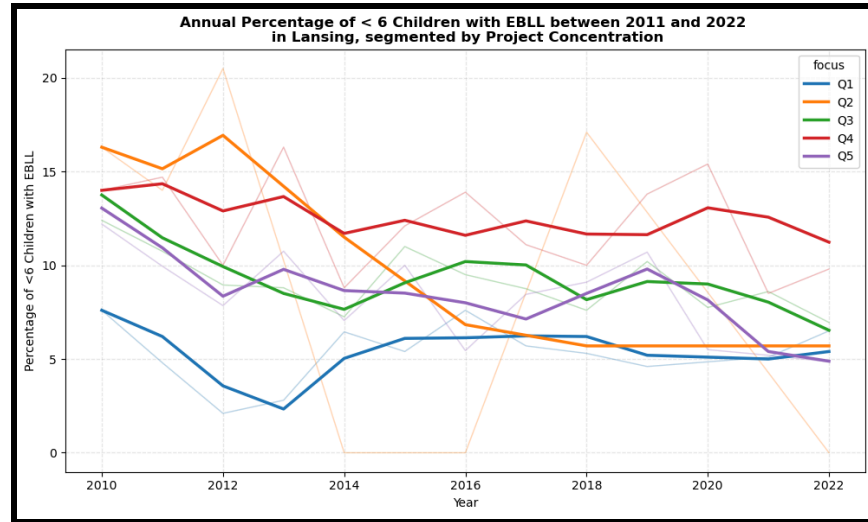


Figure 2. On average, EBLL decreases over the years, with the 4th Quintile showing insignificant decreases

V. Analysis for RQ2:

A. Methodology:

To build a longitudinal dataset for RQ2, I combined Lead Safe Lansing's administrative records with the Federal Reserve's Annual Producer Price Index (PPI) for Residential Construction (WPUIP2311001), which closely aligns with abatement activities. Using the year-level aggregation from RQ1, I aggregated total project counts and average rehabilitation costs by year into a single table, then integrated this with the annual PPI percentage changes using the year variable. Two unique visualizations were generated, the first being a dual-axis line plot comparing annual project counts versus the percentage change from 2010 to 2022, thus capturing co-movement and throughput over time, and the latter being a scatter plot with a regression line of mean rehabilitation cost as a function of annual PPI percentage change, labeled by year so as to focus on the cost-inflation relationship. To test assumptions, I used Pearson's correlation for linear associations and Spearman's rank correlation as a nonparametric alternative due to the small sample size ($n=12$) and right-skewed distribution. I also conducted weighted least squares (WLS) with HC3 robust standard errors. To ensure robustness, I removed anomalous 2021

data using the interquartile range (IQR) method and bootstrapped the model with 10,000 iterations, yielding a 95% confidence interval that supports the normality assumption.

B. Finding & Interpretation:

1. Finding:

Rehab costs do not increase with inflation, as the research question anticipates, but rather move in the opposite direction. The dual-axis line plot shows that, for much of the observed period, project counts and the percentage change in PPI move together rather than in opposite directions, which is counterintuitive; by contrast, the COVID period shows a normal relationship between the two. Both Spearman and Pearson correlation tests yield negative correlations of -0.258 and -0.433 between inflation and project counts, but the p-values exceed 0.05, so I failed to reject the null hypothesis of no significant relationship. Turning directly to rehabilitation costs, the scatter plot, which was constructed after removing the anomalous 2021 data point via IQR, reveals a statistically significant negative correlation between the annual percentage change in PPI and the mean rehabilitation cost. All four statistical tests confirm this direction: Pearson and Spearman both return significant negative correlations ($p < 0.05$), bootstrap resampling with 10,000 iterations yields a correlation of -0.742 ($p\text{-value} = 0.006$), and WLS similarly indicates a negative relationship. Despite that, an additional test found no plausible link between project counts and rehabilitation costs.

2. Interpretation:

Contrary to my expectations, higher construction inflation is associated with lower mean rehabilitation costs per project, not higher. This conflicting effect is most likely to be explained by the budget-constrained scoping effect: when prices of material increase, LSL will scope down to smaller projects with lower construction costs rather than reducing the number of projects altogether and, thus, compressing average rehabilitation project costs at each time they are rehabilitated under cost-induced inflation pressure. The lack of a substantial inflation-to-project-count relationship can also be explained by non-inflation factors, such as why

we observe low project count from 2010 to 2012, the timing of the grant cycle, and minimal state funding, not material cost, contributing to the declining number of projects, and the 2020–2022 decrease is confounded by COVID-19, which limits in-person abatement work. In response to RQ2, rehabilitation costs are negatively associated with changes in PPI inflation over time, which is not necessarily due to declining input prices but rather reflects LSL strategically limiting project size due to cost constraints. At the same time, the data may be too skewed to be meaningful for statistical testing, which makes them slightly less credible.

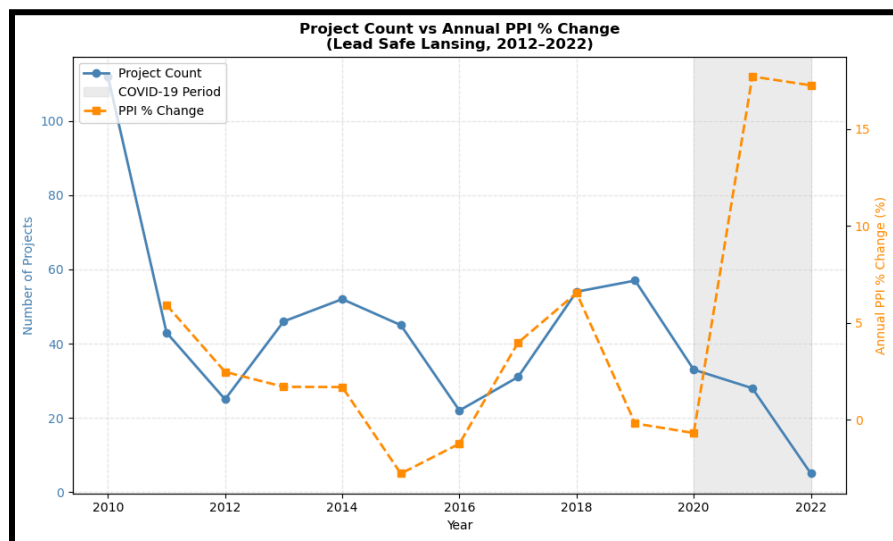


Figure 3. A dual-axis graph showing the counterintuitive relationship between the number of projects and construction material inflation, with the COVID period highlighted in grey

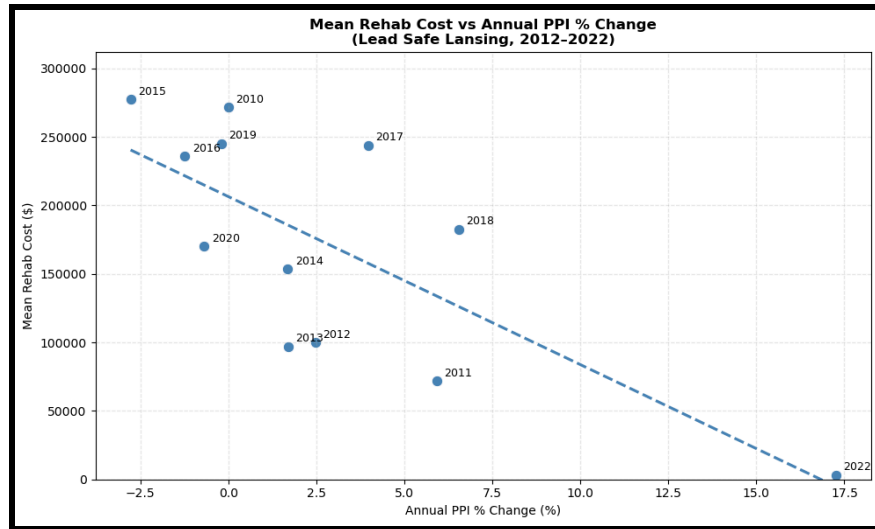


Figure 4. A scatter plot demonstrates the relationship between the annual PPI percentage change and the mean rehabilitation cost, with a downward slope regression line

VI. Conclusion: Recommendations

A. Recommendation 1: Re-focus Investment to Other Regions

1. **Audience:** Lead Safe Lansing Team & City of Lansing, which has a direct impact on the community
2. **Description:**

Upon analyzing the data, it is evident that LSL effectively targets areas with higher poverty levels, which align with increased instances of elevated blood lead levels in children. While LSL has made progress in these regions, improvement in ZIP code 48912 has been minimal despite a high concentration of projects. Inferential tests indicate persistently elevated blood lead levels in this area. I recommend conducting thorough property inspections to identify potential lead sources, such as plumbing and paint issues, and suggest employing advanced technologies, such as X-ray fluorescence (XRF). Additionally, I urge the City of Lansing to allocate resources for LSL to address the needs in ZIP code 48912, including a policy requiring targeted property inspections and prioritizing this area in future grant cycles.

B. Recommendation 2: Allocation of Capital to Abatement Projects

- 1. **Audience:** Federal and state housing and health agencies responsible for lead abatement funding (specifically HUD, MDHHS, and MSHDA) who collectively control the grant mechanisms that determine LSL's operational capacity.*
- 2. **Description:***

Through Research Question 2, I found that inflation did not significantly affect how LSL utilized its grants for projects, revealing a structural issue in funding critical initiatives by local and federal governments. To improve support for LSL's lead abatement efforts, MDHHS and MSHDA should consider reallocating funds, such as directing part of MDHHS's \$7 million in grants to Lansing rather than to cities like Detroit⁶. Increased funding could help LSL address the low project count from 2010 to 2014 and support regions with high blood lead levels in children. Additionally, HUD should pay more attention to LSL, as Lead Hazard Control Grants (LHCG) and Community Development Block Grants (CDBG) are essential for these projects⁷. I recommend that HUD annually index the spending caps for LHCG to the Producer Price Index (PPI) to counter inflation spikes that can hinder remediation efforts, especially given current global conflicts affecting prices and material availability. Lead poisoning in children can have lifelong consequences, making this issue critical.

⁶ Stebbins, Laina. "Over \$7 Million Awarded to Expand Residential Lead Hazard Services, Make Michigan Homes Safer." Health & Human Services, November 14, 2024. <https://www.michigan.gov/mdhhs/inside-mdhhs/newsroom/2024/11/14/lead-grants>.

⁷ "HUD Needs to Improve Its Award and Oversight of Lead-Based Paint Hazard Control and Lead Hazard Reduction Demonstration Grants | Oversight.gov," January 27, 2026, <https://www.oversight.gov/reports/audit/hud-needs-improve-its-award-and-oversight-lead-based-paint-hazard-control-and-lead>.